

# Limited-Display PCL Assortment Planning

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**Abstract.** Retail assortment planning converts a calibrated substitution model into a discrete display decision. Under the paired combinatorial logit model, every unordered product pair remains a choice nest when one or both products are offered. This candidate isolates that semantic requirement in a six-product, cardinality-constrained instance with a structured exact MILP and a complete enumeration oracle. The unique optimal assortment is  $\{3, 5, 6\}$ . Two independently generated programs implement different incorrect pair-state systems and both choose  $\{2, 3, 5, 6\}$ , whose correct expected margin is 11.64% below optimum. The programs solve successfully; the error lies entirely in translating the calibrated demand model.

**Keywords.** assortment optimization; paired combinatorial logit; discrete choice; revenue management; mixed-integer linear programming

Form domain: Revenue management and pricing • Model: MILP • Exact oracle: 57 feasible assortments

## 1. Business Brief

**Business-level description.** *Our category team must choose up to four products for a limited retail display. For each product, the demand-science team has estimated a customer-attraction weight and a normalized contribution margin. It has also calibrated a standard paired combinatorial logit (PCL) choice model, with a dissimilarity estimate for every unordered product pair and an outside-option weight for customers who leave without purchasing. Choose the assortment that maximizes expected contribution margin while accounting for substitution among the offered products.*

Naming the calibrated PCL engine makes the business request operationally precise while leaving the probability system and optimization formulation to the modeler.

## 2. PCL Choice System

Let  $N$  be the products,  $x_i \in \{0, 1\}$  indicate whether product  $i$  is offered,  $v_i > 0$  be attraction,  $r_i \geq 0$  be normalized margin,  $v_0 > 0$  be outside-option weight, and  $\gamma_{ij} \in (0, 1]$  be pair dissimilarity. For pair  $\{i, j\}$ , define

$$I_{ij} = \left( v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}} \right)^{\gamma_{ij}}, \quad (1)$$

$$\mu_{ij}^i = I_{ij} \frac{v_i^{1/\gamma_{ij}}}{v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}}}, \quad \mu_{ij}^j = I_{ij} - \mu_{ij}^i. \quad (2)$$

Use state binaries  $y_{ij}^{10}$ ,  $y_{ij}^{01}$ , and  $y_{ij}^{11}$  for only  $i$ , only  $j$ , and both products offered. They satisfy

$$y_{ij}^{11} \leq x_i, \quad y_{ij}^{11} \leq x_j, \quad y_{ij}^{11} \geq x_i + x_j - 1, \quad (3)$$

$$y_{ij}^{10} = x_i - y_{ij}^{11}, \quad y_{ij}^{01} = x_j - y_{ij}^{11}. \quad (4)$$

The standard PCL denominator and expected-margin numerator are

$$D(x) = v_0 + \sum_{i < j} \left( v_i y_{ij}^{10} + v_j y_{ij}^{01} + I_{ij} y_{ij}^{11} \right), \quad (5)$$

$$R(x) = \sum_{i < j} \left( r_i v_i y_{ij}^{10} + r_j v_j y_{ij}^{01} + (r_i \mu_{ij}^i + r_j \mu_{ij}^j) y_{ij}^{11} \right). \quad (6)$$

The exactly-one-offered states in Eqs. (5)–(6) are decisive: a pair vanishes only when neither member is offered.

## 3. Structured Exact MILP

The assortment problem maximizes  $R(x)/D(x)$  subject to  $\sum_i x_i \leq 4$ . Apply a Charnes–Cooper transformation with  $t = 1/D(x)$  and  $w_{ij}^s = t y_{ij}^s$  for  $s \in \{10, 01, 11\}$ . Let  $b_{ij}^s$  and  $a_{ij}^s$  denote the denominator and numerator coefficients in Eqs. (5) and (6). Valid

bounds  $0 < \underline{t} \leq t \leq \bar{t}$  follow from  $v_0 \leq D(x) \leq \bar{D}$ . The exact MILP is

$$\max \sum_{i < j} \sum_s a_{ij}^s w_{ij}^s, \quad (7)$$

$$\text{s.t. } v_0 t + \sum_{i < j} \sum_s b_{ij}^s w_{ij}^s = 1, \quad (8)$$

$$\sum_{i \in N} x_i \leq 4, \quad \text{Eqs. (3)–(4),} \quad (9)$$

$$t y_{ij}^s \leq w_{ij}^s \leq \bar{t} y_{ij}^s \quad \forall i < j, s, \quad (10)$$

$$t - \bar{t}(1 - y_{ij}^s) \leq w_{ij}^s \leq t - \underline{t}(1 - y_{ij}^s) \quad \forall i < j, s, \quad (11)$$

$$x_i, y_{ij}^s \in \{0, 1\}, \quad t, w_{ij}^s \geq 0. \quad (12)$$

Constraints (10)–(11) impose  $w_{ij}^s = t y_{ij}^s$ . The repository includes this structured Gurobi model and a separate exhaustive oracle over all 57 assortments of size at most four.

#### 4. Exact Instance and Failure Certificate

The six products are derived from the official MIT-licensed correlated synthetic instance of Feng, Che, and Chen [1]. Product IDs are remapped, and  $v_0 = 3.84538427435401$  preserves a no-purchase probability of 0.25 when all six products are displayed.

**Table 1.** Exact optimum and independently generated decisions.

| Model                                      | Selected products | Objective under that model |
|--------------------------------------------|-------------------|----------------------------|
| Correct standard PCL                       | {3, 5, 6}         | 0.323647238294198          |
| Generated singleton-plus-pair model        | {2, 3, 5, 6}      | 0.245658160940             |
| Generated jointly-offered-pairs-only model | {2, 3, 5, 6}      | 0.188961826760             |

The first generated program adds one global singleton term per selected product, instead of one exactly-one-offered contribution for every absent counterpart. The second retains a pair nest only when both products are offered. Both programs are executable and prove their own MILP optima.

**Failure certificate.** The correct PCL oracle has the unique optimum {3, 5, 6} with expected margin 0.323647238294198. Both generated programs choose {2, 3, 5, 6}. Evaluated under the calibrated standard PCL model, that decision is worth 0.285975423739680, giving absolute regret 0.037671814554518 and relative regret 11.639776%.

#### 5. Reproducible Package

The package contains the exact derivative instance, source-product mapping, structured MILP, standard-library enumeration checker, ground truth, business brief, and upstream MIT notice. The instance embeds the upstream copyright and complete MIT permission notice, so every five-file Form package carries its rights chain. The full-data no-purchase probability, probability sums, objective, bound, and direct recomputation are checked mechanically. Public package: <https://github.com/Zhengzhong-You/OR-Bench>.

#### References

- [1] J. Feng, A. Che, and Z.-L. Chen. 2026. Assortment optimization with general linear constraints under the paired combinatorial logit choice model. *INFORMS Journal on Computing*, Articles in Advance. doi:10.1287/ijoc.2024.0830.
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